GROSS ANATOMY OF TONGUE

TONGUE

Most muscles serve to attach one bone, usually via a tendon, to another. There are a few places where that is not entirely true: the ocular muscles, the scapulothoracic joint, the diaphragm and perineum are all good exceptions. However, the tongue is extraordinary. A boneless mass that you can protrude at will, fold, invert, lay flat or fill the mouth.

INTRINSIC MUSCLES

The intrinsic muscles only attach to other structures in the tongue. There are four paired intrinsic muscles of the tongue and they are named by the direction in which they travel: the superior longitudinal, inferior longitudinal, transverse and vertical muscles of the tongue. These muscles affect the shape and size of the tongue – for example, in tongue rolling – and have a role in facilitating speech, eating and swallowing. Motor innervation for the intrinsic muscles of the tongue is via the hypoglossal nerve (CNXII).

EXTRINSIC MUSCLES

Genioglossus

- Attachments: Arises from the mental symphysis and inserts into the dorsum of the tongue
- Function: Inferior fibres protrude the tongue, middle fibres depress the tongue, and superior fibres draw the tip back and down
- Innervation: Motor innervation via the hypoglossal nerve (CNXII).

Hyoglossus

- Attachments: Arises from the hyoid bone and inserts into the side of the tongue
- Function: Depresses and retracts the tongue
- Innervation: Motor innervation via the hypoglossal nerve (CNXII).

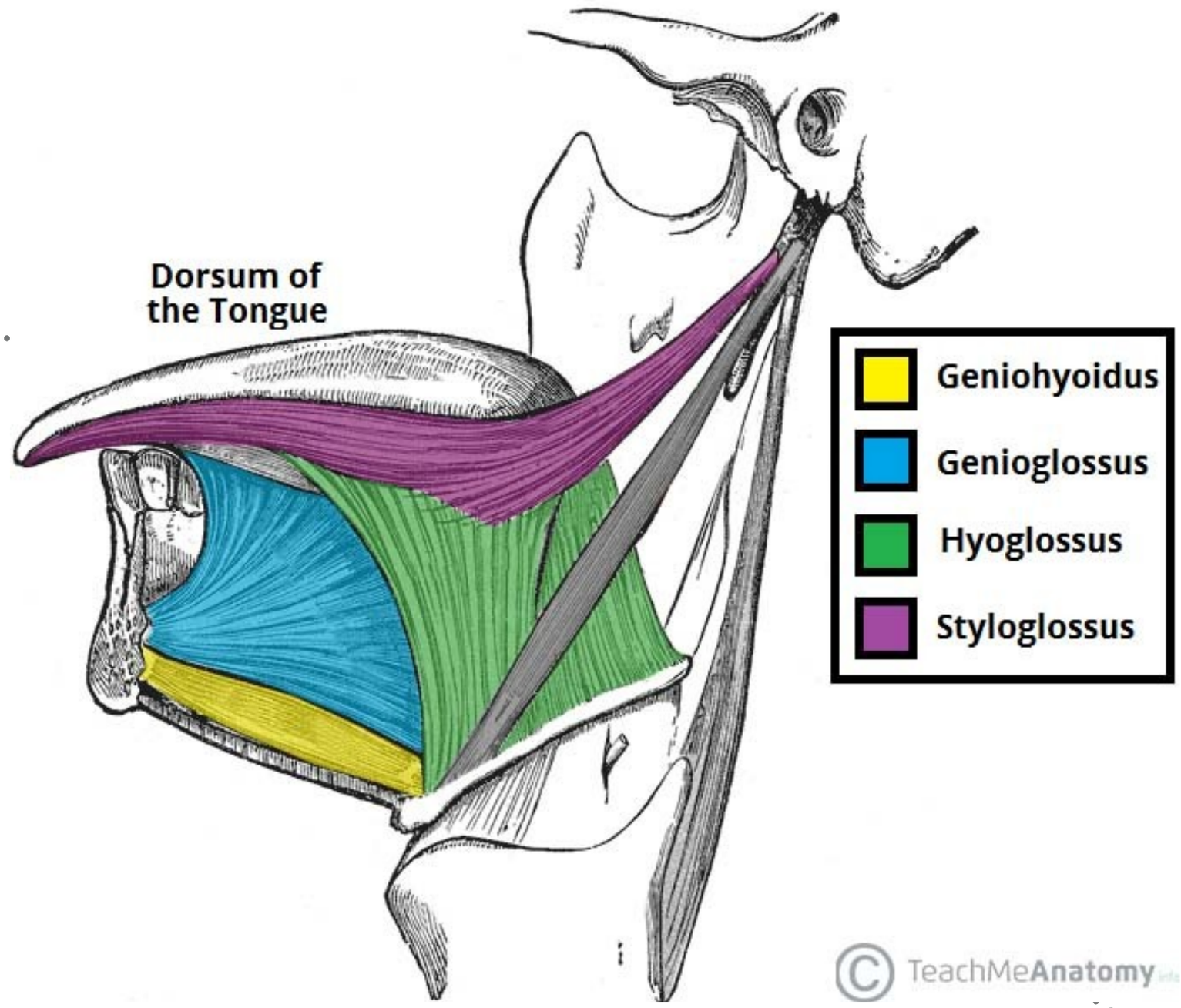
Styloglossus

- Attachments: Originates at the styloid process of the temporal bone and inserts into the side of the tongue
- Function: Retracts and elevates the tongue
- Innervation: Motor innervation via the hypoglossal nerve (CNXII).

Palatoglossus

- Attachments: Arises from the palatine aponeurosis and inserts broadly across the tongue
- Function: Elevates the posterior aspect of the tongue
- Innervation: Motor innervation via the vagus nerve (CNX).

All of the intrinsic and extrinsic muscles are innervated by the hypoglossal nerve (CN XII), except palatoglossus, which has vagal innervation (CN X).



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INNERVATION

In the anterior 2/3, general sensation is supplied by the trigeminal nerve (CNV). Specifically the lingual nerve, a branch of the mandibular nerve (CN V3). On the other hand, taste in the anterior 2/3 is supplied from the facial nerve (CNVII). In the petrous part of the temporal bone, the facial nerve gives off three branches, one of which is chorda tympani. This travels through the middle ear, and continues on to the tongue.

The posterior 1/3 of the tongue is slightly easier. Both touch and taste are supplied by the glossopharyngeal nerve (CNIX).

VASCULATURE

The **lingual artery** (branch of the external carotid) does most of the supply, but there is a branch from the facial artery, called the **tonsillar artery**, which can provide some collateral circulation. Drainage is by the **lingual vein**.

LYMPHATIC DRAINAGE

The lymphatic drainage of the tongue is as follows:

- Anterior two thirds – initially into the submental and submandibular nodes, which empty into the deep cervical lymph nodes
- Posterior third – directly into the deep cervical lymph nodes